

# Hallucination Detection Report

AI Output Verification · Knowledge Graph Analysis

Generated: 2026-04-17 10:35  
Triples: 10 · Verdicts: 4 · Inconsistencies: 0 ·  
Violations: 6

## EXECUTIVE SUMMARY

|                                 |                                            |                                         |
|---------------------------------|--------------------------------------------|-----------------------------------------|
| <div>2</div> <div>Correct</div> | <div>1</div> <div>Predicate Mismatch</div> | <div>1</div> <div>Fabricated Fact</div> |
|---------------------------------|--------------------------------------------|-----------------------------------------|

## EXTRACTED TRIPLES

All verifiable triples extracted from the LLM response before canonicalization or checking.

| Subject | Predicate         | Object      | Verifiable | Verification Type | Predicate Definition                                                                  |
|---------|-------------------|-------------|------------|-------------------|---------------------------------------------------------------------------------------|
| 575     | HAS_INTEREST_RATE | 12.21%      | Yes        | ENTITY_ATTRIBUTE  | Indicates the interest rate associated with the subject entity.                       |
| 580     | HAS_INTEREST_RATE | 11.91%      | Yes        | ENTITY_ATTRIBUTE  | Indicates the interest rate associated with the subject entity.                       |
| 579     | HAS_INTEREST_RATE | 12.03%      | Yes        | ENTITY_ATTRIBUTE  | Indicates the interest rate associated with the subject entity.                       |
| 579     | HAS_RISK_TIER     | subprime    | Yes        | ENTITY_ATTRIBUTE  | Indicates the risk classification or category assigned to the subject entity.         |
| 890     | HAS_INTEREST_RATE | Below 4.36% | Yes        | ENTITY_ATTRIBUTE  | Indicates the upper bound interest rate associated with the subject entity.           |
| 890     | HAS_INTEREST_RATE | ~4.0%       | Yes        | ENTITY_ATTRIBUTE  | Indicates the estimated interest rate associated with the subject entity.             |
| 890     | HAS_RISK_TIER     | super prime | Yes        | ENTITY_ATTRIBUTE  | Indicates the risk classification or category assigned to the subject entity.         |
| 890     | HAS_DOWN_PAYMENT  | Below 2.0%  | Yes        | ENTITY_ATTRIBUTE  | Indicates the upper bound down payment amount associated with the subject entity.     |
| 890     | HAS_DOWN_PAYMENT  | ~1.5%       | Yes        | ENTITY_ATTRIBUTE  | Indicates the estimated down payment amount associated with the subject entity.       |
| 335     | HAS_LOAN_TERM     | 12 months   | Yes        | ENTITY_ATTRIBUTE  | Indicates the duration or term length of the loan associated with the subject entity. |

## CANONICALIZED TRIPLES

Predicates normalized against the database schema. Shows original predicate, canonical match, and match confidence.

| Subject | Original Predicate | Object      | Canonical Predicate                                       | Confidence |
|---------|--------------------|-------------|-----------------------------------------------------------|------------|
| 575     | HAS_INTEREST_RATE  | 12.21%      | INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE | HIGH       |
| 580     | HAS_INTEREST_RATE  | 11.91%      | INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE | HIGH       |
| 579     | HAS_INTEREST_RATE  | 12.03%      | INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE | HIGH       |
| 579     | HAS_RISK_TIER      | subprime    | RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL             | HIGH       |
| 890     | HAS_INTEREST_RATE  | Below 4.36% | INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE | HIGH       |

| Subject | Original Predicate | Object      | Canonical Predicate                                         | Confidence |
|---------|--------------------|-------------|-------------------------------------------------------------|------------|
| 890     | HAS_INTEREST_RATE  | ~4.0%       | INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE   | MEDIUM     |
| 890     | HAS_RISK_TIER      | super prime | RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL               | HIGH       |
| 890     | HAS_DOWN_PAYMENT   | Below 2.0%  | DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION | HIGH       |
| 890     | HAS_DOWN_PAYMENT   | ~1.5%       | DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION | MEDIUM     |
| 335     | HAS_LOAN_TERM      | 12 months   | LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED       | HIGH       |

### LOGICAL INCONSISTENCIES

Contradictions detected between triples within the LLM response, independent of the database.

No logical inconsistencies detected.

### SYMBOLIC RULE VIOLATIONS

Violations caught by explicit domain constraints before database lookup. Violated triples were removed from the evidence retrieval step.

#### c\_tier\_300\_579 • • MEDIUM

579 → RISK\_TIER\_CLASSIFIED\_AS\_BORROWER\_DANGER\_LEVEL → subprime

Credit score 579 is within 300-579, but risk tier is 'subprime' instead of required 'deep subprime'.

#### c\_interest\_rate\_range • • MEDIUM

890 → INTEREST\_RATE\_CHARGED\_AS\_ANNUAL\_BORROWING\_COST\_PERCENTAGE → Below 4.36%

Credit score 890 > 850, which violates limit regardless of interest rate.

#### c\_interest\_rate\_range • • MEDIUM

890 → INTEREST\_RATE\_CHARGED\_AS\_ANNUAL\_BORROWING\_COST\_PERCENTAGE → ~4.0% (estimated)

Credit score 890 > 850, which violates limit regardless of interest rate.

#### c\_risk\_outside\_range • • MEDIUM

890 → RISK\_TIER\_CLASSIFIED\_AS\_BORROWER\_DANGER\_LEVEL → super prime

Credit score 890 > 850 is outside allowable range; any subject outside 300-850 violates.

#### c\_down\_payment\_range • • MEDIUM

890 → DOWN\_PAYMENT\_REQUIRED\_AS\_UPFRONT\_BORROWER\_CASH\_CONTRIBUTION → Below 2.0%

Credit score 890 > 850, which violates range regardless of down payment amount.

**c\_down\_payment\_range** · · **MEDIUM**

890 → DOWN\_PAYMENT\_REQUIRED\_AS\_UPFRONT\_BORROWER\_CASH\_CONTRIBUTION → ~1.5% (estimated)

Credit score 890 > 850, which violates range regardless of down payment amount.

## DETECTED HALLUCINATIONS

| Subject | Predicate                                                 | Object    | Verdict            | Match  | Violated  | Conf.                                                                                     | Explanation                                                                                                                                                                                                                                                                                                                                                           |
|---------|-----------------------------------------------------------|-----------|--------------------|--------|-----------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 575     | INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE | 12.21%    | Correct            | STRICT | —         | <div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> </div> 1.00 | <p>Object value '12.21%' semantically matches DB entry '12.21' as the number is identical and the percent symbol does not affect the meaning.</p> <p>Evidence: subject: 575, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 12.21</p>                                                                                                  |
| 580     | INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE | 11.91%    | Correct            | STRICT | —         | <div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> </div> 1.00 | <p>Object value '11.91%' semantically matches DB entry '11.91' as the number is identical and the percent symbol does not affect the meaning.</p> <p>Evidence: subject: 580, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 11.91</p>                                                                                                  |
| 579     | INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE | 12.03%    | Fabricated Fact    | NONE   | SUBJECT   | <div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> </div> 0.00 | <p>Neither subject nor object appears in any triple with this predicate, indicating the fact is fabricated.</p>                                                                                                                                                                                                                                                       |
| 335     | LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED     | 12 months | Predicate Mismatch | NONE   | PREDICATE | <div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> </div> 0.60 | <p>Subject '335' exists but does not have the predicate 'LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED' among its available predicates.</p> <p>Evidence: subject: 335, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 335, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 20.88</p> |

TRIPLE AUDIT TRAIL — ORIGINAL → CANONICAL → VERDICT

Full pipeline trace for every triple. UNVERIFIABLE\* means the triple had no verdict — e.g. dropped due to output truncation.

| Orig. Subject | Orig. Predicate   | Orig. Object | Confidence | Canon. Predicate                                          | Canon. Object | Verdict            |
|---------------|-------------------|--------------|------------|-----------------------------------------------------------|---------------|--------------------|
| 575           | HAS_INTEREST_RATE | 12.21%       | HIGH       | INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE | 12.21%        | CORRECT            |
| 580           | HAS_INTEREST_RATE | 11.91%       | HIGH       | INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE | 11.91%        | CORRECT            |
| 579           | HAS_INTEREST_RATE | 12.03%       | HIGH       | INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE | 12.03%        | FABRICATED_FACT    |
| 335           | HAS_LOAN_TERM     | 12 months    | HIGH       | LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED     | 12 months     | PREDICATE_MISMATCH |

VERDICT LEGEND

| Verdict              | Description                                                            |
|----------------------|------------------------------------------------------------------------|
| Correct              | All three components (subject, predicate, object) verified against DB. |
| Rule Violation       | One or more components contradict DB evidence.                         |
| Multi-Rule Violation | Violation spans multiple DB triples simultaneously.                    |
| Unverifiable         | No usable DB evidence found for any component.                         |
| Predicate Mismatch   | Subject real, but stated predicate is not a valid relation for it.     |
| Fabricated Fact      | Triple asserts a relationship that does not exist anywhere in DB.      |